UCIP



URBAN CLIMATE INTELLIGENCE PLATFORM

# Mumbai's ward-level heat vulnerability, turned into decisions you can act on.

A decision-support tool that tells Mumbai's city planners which of the 24 BMC wards to cool first, why, with what intervention, and where a fixed budget goes.

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| GRADE | 11 (IBDP Year 1) |
| SELECTED DOMAIN | PLANET |
| FOCUS AREA | Urban heat resilience / climate adaptation decision-support |

## What problem are you solving?

Mumbai is heating unevenly. Heat pockets form over slums, barren land, and impervious concrete, but the city has no ward-level, evidence-based way to decide where to invest cooling infrastructure first. Current plans set city-wide goals without per-ward targeting.

## Who is affected?

The most heat-vulnerable residents: slum households (heat-trapping roofs, low cooling access), the elderly, and wards far from hospitals. Vulnerability is highest exactly where data is scarcest.

## Why does this problem matter?

Heat is India's deadliest weather hazard, and Ahmedabad's Heat Action Plan proved targeted city action saves roughly 1,190 lives a year (Knowlton et al. 2014). Mumbai's own climate plan (MCAP 2022) names heat as a priority pillar but stops at city-level strategy: goals without a reproducible method for choosing which ward gets cooled first.

## Where does this problem exist?

Every dense tropical city; UCIP scopes it tightly to Mumbai's 24 BMC wards, where IIT Bombay remote-sensing studies already show strong surface urban-heat-island effects over informal housing (Mehrotra, Bardhan and Ramamritham 2018).

## Data, facts, observations

- Live satellite pipeline over Mumbai (Google Earth Engine): 12 Landsat 8/9 dry-season scenes, mean land-surface temperature 33.6 °C, mean NDVI 0.305, computed, not copied.
- WorldPop age-sex rasters (2020, most recent year available for India) give elderly population per ward; OSM gives hospital locations; mapped Datameet slum-cluster boundaries proxy the slum index.
- Tree canopy cooling is nonlinear: below ~40% canopy, daytime cooling is negligible; 40 to 80% gives approximately 1 °C (Ziter et al. 2019, PNAS). Cool roofs cut peak temperature ~0.6 K per +0.1 albedo, conservative end of the published range (Santamouris 2014).

## Existing solutions explored (5 closest)

| Tool | Gap |
|---|---|
| Mumbai Climate Action Plan 2022 | Strategy document, no per-ward ranking method |
| RAND / Azhar et al. 2017 India HVI | District-level (640 districts), too coarse for one city; maps vulnerability then stops |
| IIT Bombay Mumbai UHI studies | Rigorous heat maps, but diagnostic only |
| C40 Urban Cooling Toolbox | Real decision-support, but global/generic, no Mumbai data under it |
| Ahmedabad Heat Action Plan | Temporal early-warning (when heat strikes), not spatial investment targeting (where to build) |

## Gaps discovered

Every prior tool does at most two of five things: ward-level ranking, per-factor explainability, an intervention engine, an ecological guardrail, a budget layer. None does all five. Most tools also overstate certainty; none openly discloses proxy-data limitations.

## The idea

UCIP computes a Heat Vulnerability Index (HVI) for each of Mumbai's 24 wards from satellite and demographic data, with PCA-derived weights following the peer-reviewed Reid et al. 2009 methodology (data-driven, not guessed). Every ward gets a transparent per-factor breakdown (no black box), and the index is sensitivity-tested (weights perturbed +/-20% to show the ranking holds).

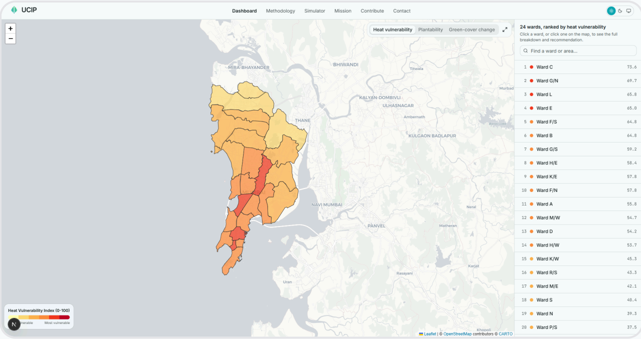
## How it solves the problem

The index feeds a rule-based Nature-Based-Solutions engine where every recommendation carries a citation, gated by an ecological plantability filter: native afforestation only where restoration is appropriate (Bastin et al. 2019), blocked over native grasslands where planting would backfire (Veldman et al. 2019), routing those areas to cool roofs instead. Recommendations roll up into ward cards a planner can act on. Not another "plant trees everywhere" map.

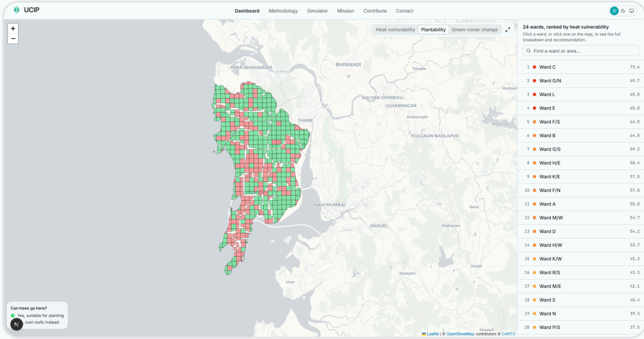
## Who benefits

- BMC planners: a defensible, explainable priority list instead of intuition
- Vulnerable residents: cooling investment lands in the highest-need wards first
- Researchers and NGOs: open methodology, open data sources, stated limitations

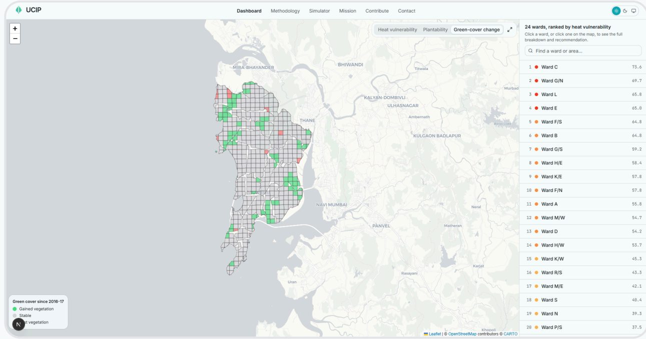
# 05. PROTOTYPE / SOLUTION DESIGN



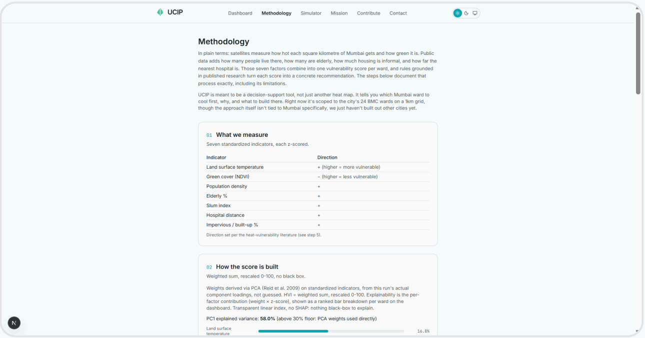
Leaflet choropleth of the 24-ward HVI ranking, plus ward cards (rank, factor contribution bars, recommended intervention)



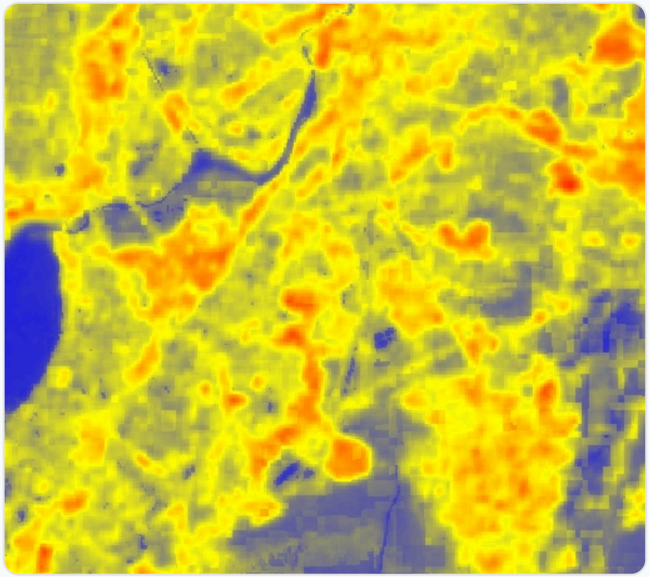
NBS plantability filter: green cells eligible for afforestation, red cells routed to cool roofs instead



Green-cover change layer: gained / stable / lost NDVI classification per cell



Methodology page: real computed weights, sensitivity chart, citations, stated limitations



Pipeline evidence: GEE land-surface-temperature composite (Day-0 spike)

Pipeline evidence: GEE NDVI composite (Day-0 spike)

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Stack: Google Earth Engine (Landsat 8/9, WorldPop, WorldCover) → Python (geopandas, scikit-learn PCA) → Supabase + PostGIS → Next.js + Leaflet, deployed on Vercel.
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## Expected impact

Turns Mumbai's existing heat diagnosis into an operational bridge: which ward first, why, what intervention, at what cost. Methodology is reproducible for any Indian city with the same open data (all sources are free: Landsat, WorldPop, OSM, GHSL). Honesty layer builds trust: proxy data disclosed, land-surface temperature stated as not equal to air temperature, coefficients marked as transferred, not Mumbai-calibrated.

## How it can be improved or expanded

- Finer grid (1 km to 500 m) and Census-linked demographic data when accessible
- NDVI change detection over time to verify interventions actually green the ward
- What-if simulator: slide canopy % / cool-roof fraction, watch predicted LST fall (stretch)
- Budget optimizer: maximize vulnerability-reduction per rupee across wards (stretch)
- Replication kit: config-driven port to Pune, Ahmedabad, Delhi wards